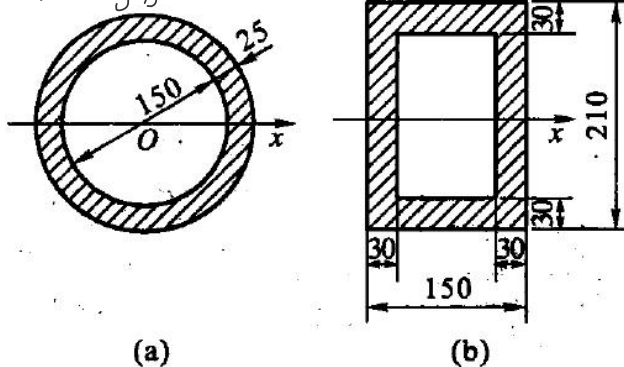


5-3 试分别求图示环形和箱形截面对其对称轴  $x$  的惯性矩。

$$(a) I_x = \int_A y^2 dA = \int_{75}^{100} \left( \int_0^{2\pi} x^2 \sin^2 x d\theta dx \right) = \int_{75}^{100} \pi x^2 dx = \frac{\pi}{4} (100^4 - 75^4) \approx 5.37 \times 10^7 \text{ mm}^4$$

$$(b) I_x = \frac{bh^3}{12} - \frac{bh^3}{12} = \frac{150 \times 210^3}{12} - \frac{90 \times 150^3}{12} \approx 9.05 \times 10^7 \text{ mm}^4$$



题 5-3 图

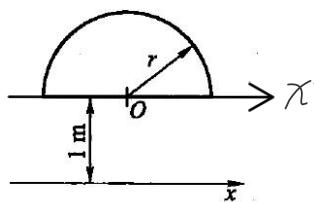
5-5 试求图示  $r=1 \text{ m}$  的半圆形截面对其  $x$  轴的惯性矩, 其中  $x$  轴与半圆形的底边平行, 相距  $1 \text{ m}$ 。

由平行轴定理

$$I_{x'} = I_{xc} + \frac{1}{2} \pi r^2 \cdot \frac{4r}{3\pi} = \frac{1}{2} I_{\text{圆}}$$

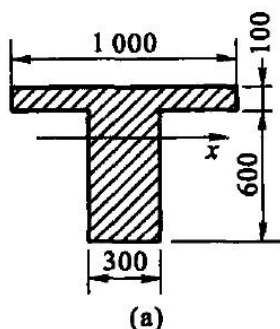
$$I_x = I_{xc} + \frac{1}{2} \pi r^2 \left( 1 + \frac{4r}{3\pi} \right)$$

$$\Rightarrow I_x = \left( \frac{4r}{3\pi} \right)^2 \cdot \frac{\pi r^2}{2} + \left( \frac{4r}{3\pi} + 1 \right)^2 \cdot \frac{\pi r^2}{2} \approx 3.3 \text{ m}^4$$

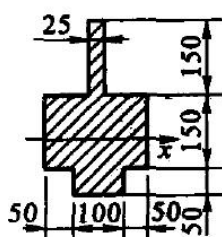


题 5-5 图

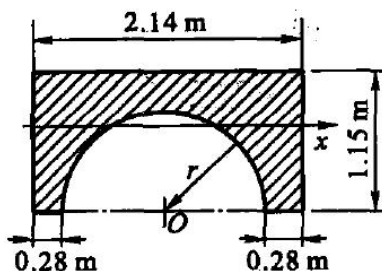
5-8 试求下列各截面对其形心轴  $x$  的惯性矩。



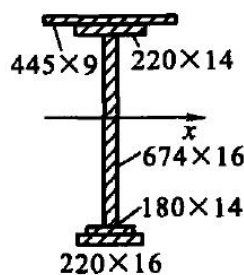
(a)



(b)



(c)



(d)

(a) 先求形心:

$$y_c = \frac{300 \times 600 \times 300 + 1000 \times 100 \times (600 + 50)}{300 \times 600 + 1000 \times 100} = 425$$



$$I_z = I_{z1} + I_{z2}$$

$$= \frac{1}{12} 300 \times 600^3 + 300 \times 600 \times (425 - 300)^2 + \frac{1}{12} 1000 \times 100^3 + 1000 \times 100 \times (650 - 425)^2 \approx 1.336 \times 10^9 \text{ mm}^4$$

(b) 先求形心

$$y = \frac{100 \times 50 \times 25 + 200 \times 150 \times 125 + 275 \times 25 \times 150}{100 \times 50 + 200 \times 150 + 275 \times 25} = \frac{490625}{3875} = \frac{3925}{31}$$

$$I_z = I_{z1} + I_{z2} + I_{z3} = \frac{1}{12} \times 25 \times 150^3 + \frac{1}{12} \times 200 \times 150^3 + \frac{1}{12} \times 100 \times 50^3 + 25 \times 150 \times \left(25 - \frac{3925}{31}\right)^2 + 200 \times 150 \times \left(\frac{3925}{31} - 125\right)^2 + 100 \times 50 \times \left(\frac{3925}{31} - 25\right)^2 \approx 1.987 \times 10^8 \text{ mm}^4$$

(c) 先求形心

$$y_c = \frac{2.14 \times 1.15 \times \frac{1.15}{2} - \frac{1}{2} \pi \times 0.79^2 \times \frac{4 \times 0.79}{3\pi}}{2.14 \times 1.15 - \frac{1}{2} \pi \times 0.79^2}$$

$$I_z = I_{矩} - I_{半圆} = \frac{1}{12} \times 2.14 \times 1.15^3 + 2.14 \times 1.15 \times \left(\frac{1.15}{2} - y_c\right)^2 - \left[ \frac{1}{128} \pi \times 1.58^4 + \frac{1}{2} \pi \times 0.79^2 \times \left(\frac{4 \times 0.79}{3\pi} - y_c\right)^2 \right] \approx 1.34 \times 10^4 \text{ mm}^4$$

(d) 先求形心

$$y_c = \frac{220 \times 16 \times 8 + 180 \times 14 \times 23 + 16 \times 674 \times \left(\frac{674}{2} + 30\right) + 220 \times 14 \times (30 + 674 + 7) + 9 \times 445 \times \left(\frac{9}{2} + 14 + 674 + 30\right)}{220 \times 16 + 180 \times 14 + 16 \times 674 + 445 \times 9 + 220 \times 14} = 381.75 \text{ mm}$$

$$I_x = I_{x1} + I_{x2} + I_{x3} + I_{x4}$$

$$= \frac{1}{12} \times 220 \times 16^3 + \frac{1}{12} \times 180 \times 14^3 + \frac{1}{12} \times 220 \times 14^3 + \frac{1}{12} \times 445 \times 9^3 + \frac{1}{12} \times 16 \times 674^3 + 220 \times 16 \times (8 - y_c)^2 + 180 \times 14 \times (y_c - 23)^2 + 674 \times 16 \times \left(y_c - \frac{674}{2} - 30\right)^2 + 220 \times 14 \times (y_c - 674 - 30 - 7)^2 + 445 \times 9 \times \left(y_c - 674 - 30 - 14 - \frac{9}{2}\right)^2$$

$$\approx 2.03 \times 10^9 \text{ mm}^4$$

